

Executive Teaser

Evolutionary Trajectory Fine-Tuning · Autonomous SOTA+1 Discovery

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The Problem — Why R&D is Broken

Every year, billions of dollars are poured into R&D cycles that move at the speed of human cognitive limits. AI models today generate code — but they do not *invent*. They retrieve patterns from frozen, outdated weights. The result: every state-of-the-art architecture was discovered by a human, not a machine. **The next breakthrough is always sitting in a "Knowledge Blindspot" the model cannot see.**

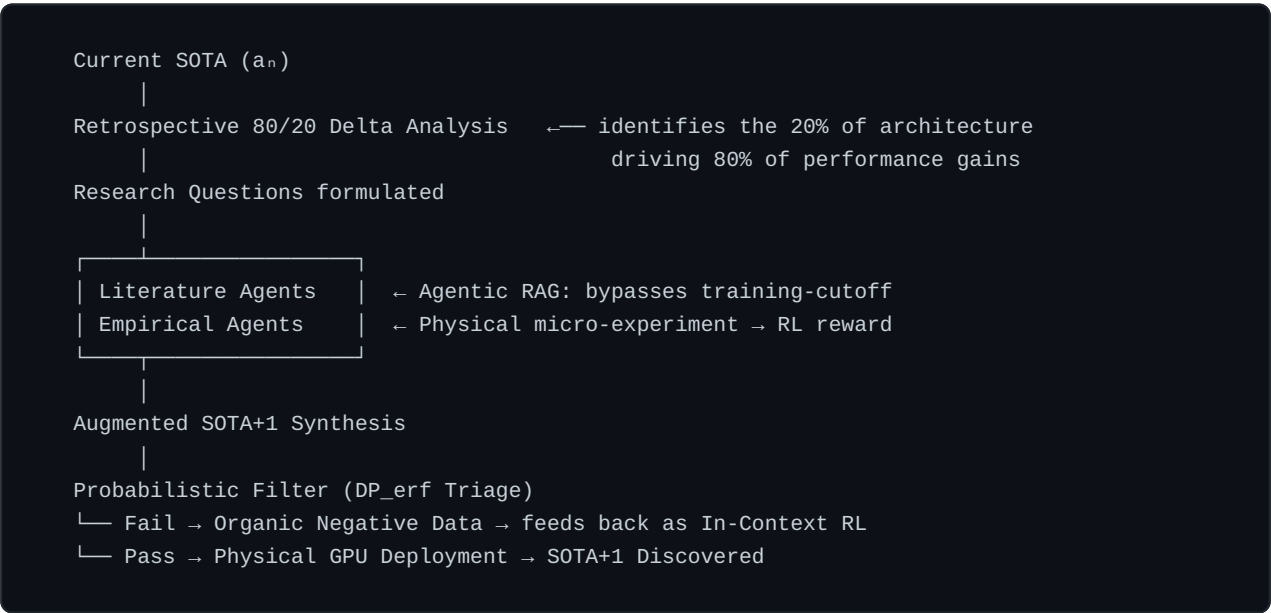
The Solution — ETFT: Teaching Machines *How* to Invent

Evolutionary Trajectory Fine-Tuning (ETFT) is a novel ML paradigm that transforms LLMs from static code generators into **autonomous research scientists**.

Instead of training on disconnected code snippets, ETFT teaches models the *evolutionary trajectory* of optimization — the precise, AI-verified sequence of architectural jumps ($a_0 \rightarrow a_1 \rightarrow \dots \rightarrow a_n$) that produced every SOTA algorithm ever written.

The system does not predict the next token. It predicts the next epoch.

The Open-Loop Empirical Architecture closes the Knowledge Blindspot by deploying autonomous Literature Agents (real-time arXiv retrieval) and Empirical Agents (micro-experiments with RL feedback) — continuously recycling even *failed* hypotheses as high-value training signal.



(Figure 2 — Open-Loop Empirical ETFT Ecosystem)

The Unfair Advantage — Three Moats in One

Moat	What it means
Recursive negative data	Failed experiments are <i>not wasted</i> — they harden the Probabilistic Heuristic Filter, creating a self-improving dataset no competitor can buy.
Pareto Delta Analysis	Deterministic 80/20 analysis of historical deltas (a_{n-1} vs a_n) teaches the model <i>where</i> breakthroughs hide — before any inference is run.
Domain-agnostic architecture	The same pipeline that invents the next neural network architecture can evolve a pharmaceutical molecule, a supply-chain protocol, or a foundational scientific paper.

Validated — Without a Million-Dollar Cluster

The "GPU-Poor" PoC (April 2026) demonstrates autonomous SOTA+1 discovery in TinyML time-series classification using only cloud reasoning APIs + a consumer RTX 4090:

- Evolutionary trajectory injected into context: MLP → 1D CNN → Depthwise CNN → MobileNetV2
- Autonomous agent identified 1D State Space Models (Mamba-style) from literature published *after* training cutoff
- Result: **Accuracy 94.2% · Peak RAM 11 KB** — surpassing human-designed SOTA (93%, 18 KB)
- Cost: **minutes, not months**

The Vision — General Innovation Accelerator

ETFT does not stop at deep learning. Any process that can be described as an ordered sequence of improvements — from materials science to drug discovery to organizational design — becomes a target. We are building the **operating system for autonomous scientific progress**.

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